

VENKATESH MUDALIAR

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SUMMARY

AI Engineer with hands-on experience building agentic AI systems, RAG pipelines, and LLM-powered applications, backed by 4+ years of production engineering experience across Python, APIs, data, and cloud/enterprise systems. M.S. in Data Science from Stevens Institute of Technology (GPA 3.82).

TECHNICAL SKILLS

AI/ML & Agentic AI: Multi-Agent Systems (LangGraph), RAG (LangChain, FAISS, ChromaDB), LLM Fine-Tuning (LoRA, QLoRA), Prompt Engineering, PyTorch, TensorFlow, Scikit-learn

Languages & Backend: Python, Java, SQL, Node.js

Databases: PostgreSQL, DynamoDB, Snowflake

Cloud & DevOps: AWS (Bedrock, Lambda, SageMaker, S3, EC2, IAM, CloudFormation), GCP, Docker, CI/CD

GenAI APIs & Tools: Anthropic Claude API, AWS Bedrock (Nova), FastAPI, pytest

Enterprise Integration: SAP CPI, SAP BTP, PI/PO, S/4HANA, OAuth 2.0, SAML

PROFESSIONAL EXPERIENCE

Software Engineer (Intern) | Easley Dunn Productions

Jun 2026 - Present

- Reverse-engineered data models across two production codebases (**300+ files**) to build a verified schema-mapping document, surfacing 5+ duplicate **data pipelines** and identifier inconsistencies for engineering leadership.
- Automated** static code audits across two large codebases using **LLM-based** analysis tooling, extracting configuration sources and probability models into an evidence-cited technical report in days instead of weeks.

ML Research Assistant | Stevens Institute of Technology, Hoboken, NJ

Oct 2024 - May 2026

- Designed ML evaluation pipelines benchmarking traditional models against **LLM-based forecasting** on financial datasets, improving F1-score **12%** via cross-validated feature selection.
- Built **FAISS**-based semantic search and retrieval-augmented pipelines integrating multi-source financial data, improving retrieval precision **18%** across LLM benchmarking experiments.

Integration Engineer | Accenture, India

Feb 2023 - Sep 2024

- Built cloud-based API and integration pipelines connecting SAP S/4HANA with Salesforce, SuccessFactors, Ariba, and external systems using REST, OData, SOAP, and IDoc.
- Spearheaded a large-scale ERP migration, modernizing **50+ SAP integration interfaces** from legacy PI/PO middleware to SAP's cloud-native CPI platform, cutting data-processing time **30%**.
- Refactored 30+ legacy interfaces into modular, reusable CPI pipelines, improving maintainability and driving a **30% throughput gain**.

Integration Engineer | LTIMindtree Ltd., India

Sep 2020 - Feb 2023

- Engineered custom Groovy scripts to catch and **handle errors** in high-volume OData batch processing, cutting manual troubleshooting and improving integration reliability.
- Led end-to-end automation of Sourcing and Procurement workflows in SAP S/4HANA Cloud, replacing manual processes and cutting manual intervention **60%**.
- Translated business and client requirements into technical specifications for new SAP integrations, performing gap analysis to guide the Canada **migration**.

AI/ML PROJECTS

SAP IntelliOps: Multi-Agent Incident Resolution | [GitHub](#)

- Built an agentic incident-resolution system that investigates SAP CPI failures using LangGraph-orchestrated agents, RAG, and tool calling, with policy-gated execution and human approval for higher-risk remediation.
- Evaluated the system across incident classification, remediation, and escalation scenarios, achieving 20/20 taxonomy accuracy, 16/16 known-remediation accuracy, and 1.00 escalation F1.

RagProbe: Adversarial LLM & RAG Security Benchmarking | [GitHub](#)

- Built an adversarial **RAG** evaluation framework (FAISS, LangChain, LLM-as-a-judge) benchmarking hallucination, prompt injection, and faithfulness across 500+ test cases.
- Used the evaluation results to reduce hallucination rate **22%** and improve context recall **18%** across the benchmarked pipeline.

EDUCATION

M.S. Data Science

Stevens Institute of Technology | GPA 3.82

Sep 2024 - May 2026

Hoboken, NJ

B.E. Electronics & Telecommunications

University of Mumbai

Apr 2016 - May 2020

Mumbai, India

CERTIFICATIONS & AWARDS

AWS Certified AI Practitioner | IBM Data Science Professional Certificate | Databricks Generative AI Fundamentals

Awards: GoMax Award, LTIMindtree | Best Performer of the Month, Accenture